Research Seminar

Group N

11. FLOW TOXICITY AND LIQUIDITY IN A HIGH-FREQUENCY WORLD

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Review of Financial Studies 25(5), 2012

Name Surname · Name Surname · Name Surname

Why toxic order flow matters

- **Market makers** (today mostly high-frequency firms) earn a tiny spread on many trades and avoid directional bets.
- They lose when the other side **knows more**. Order flow is **toxic** when it adversely selects them, often before they notice.
- When toxicity rises they widen quotes or **leave**. With no one absorbing orders, prices gap. The paper links the **6 May 2010 flash crash** to this.
- **Question:** can toxicity be measured in real time, from trades alone, early enough to act?



Method: from PIN to VPIN, measured in volume time

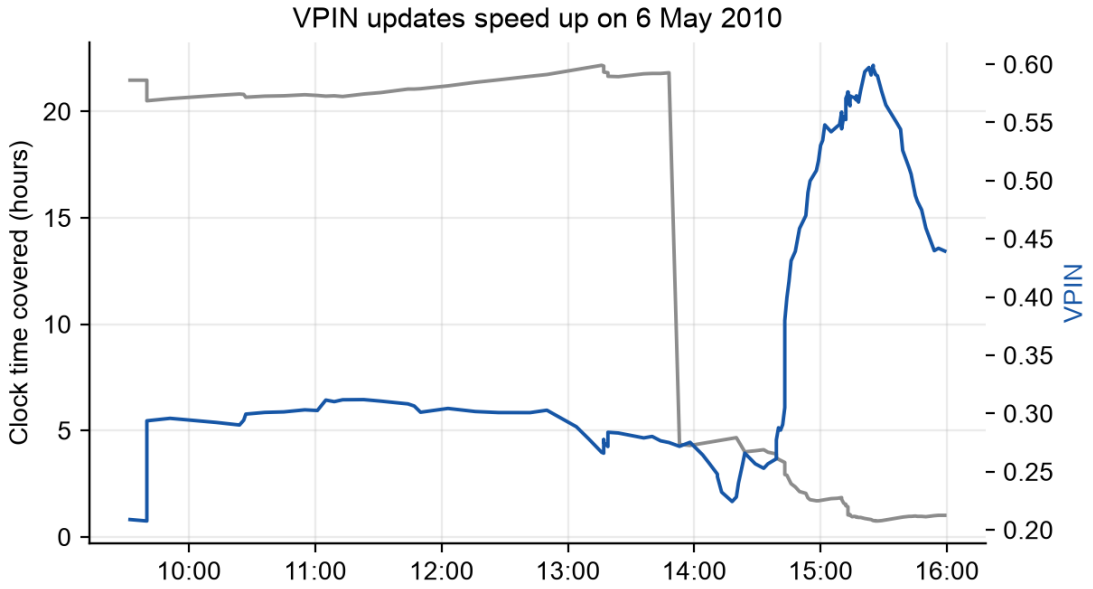
• **PIN** (Easley et al. 1996): the share of orders that come from informed traders

$$\mathrm{PIN} = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}$$

• α: chance of news, μ: informed arrivals, ε: uninformed arrivals per side. Needs maximum likelihood: slow and unstable at high frequency.

• **VPIN:** |buys − sells| estimates αμ, total volume estimates αμ + 2ε. No parameters to fit.

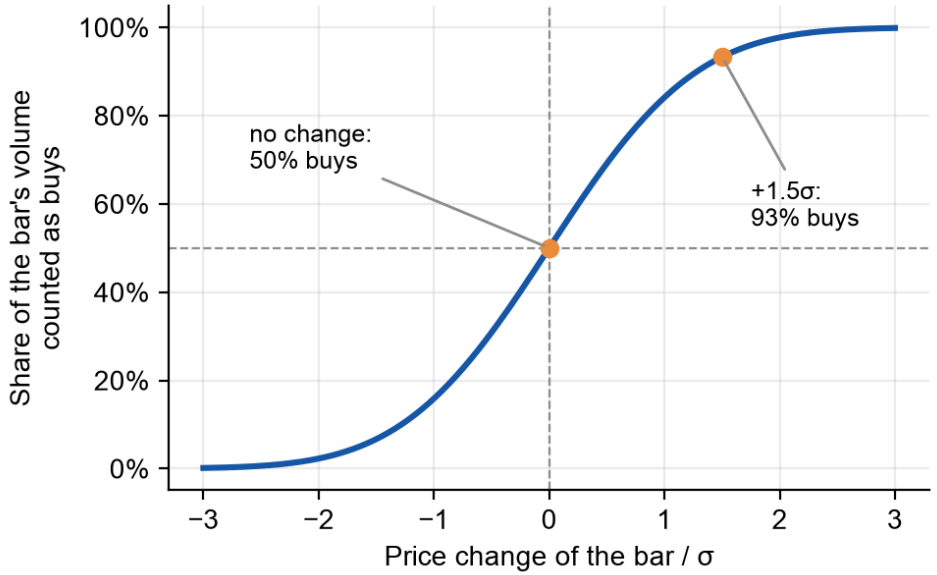
• **Volume clock:** cut trading into buckets of equal volume (1/50 of a day). Busy, news-heavy periods get more updates.



Method: bulk volume classification

$$V_\tau^B = \sum_{i \in \tau} V_i Z\left(\frac{P_i - P_{i-1}}{\sigma_{\Delta P}}\right), \quad V_\tau^S = V - V_\tau^B$$

- Group trades into **one-minute bars**; split each bar's volume by its standardized price change.
- No price change → 50/50. A rise of 1.5σ → 93% buys. Z is the standard normal CDF; σ is the standard deviation of bar price changes.
- Robust to order splitting and cancellations, cheap to compute.
- **But:** the imbalance it produces grows with |price change|, by construction.



What is estimated, what is chosen, what is evaluated

Step	Objective	Data used
Estimate PIN (original model)	Maximise the likelihood of a Poisson mixture: α, μ, ε, δ	daily buy and sell counts
Estimate VPIN	Nothing is fitted: average |sell − buy volume| / V, a moment estimate of αμ / (αμ + 2ε)	the last n buckets
Inputs σ and V	Standard deviation of one-minute price changes; average daily volume / 50	paper: whole sample · ours, real-time: previous 250 days
Choose buckets/day and n	Paper: grid cell with the highest corr(ln VPIN, next |return|) (Fig. 10) or true-positive rate (Table 5)	paper: the evaluation sample itself · ours: 2009–11, then frozen
Evaluate	Corr with next |return|; P(VPIN high | big move); true positives. We add lift and ΔR^2 over volatility	paper: in sample · ours: 2011–26, out of sample

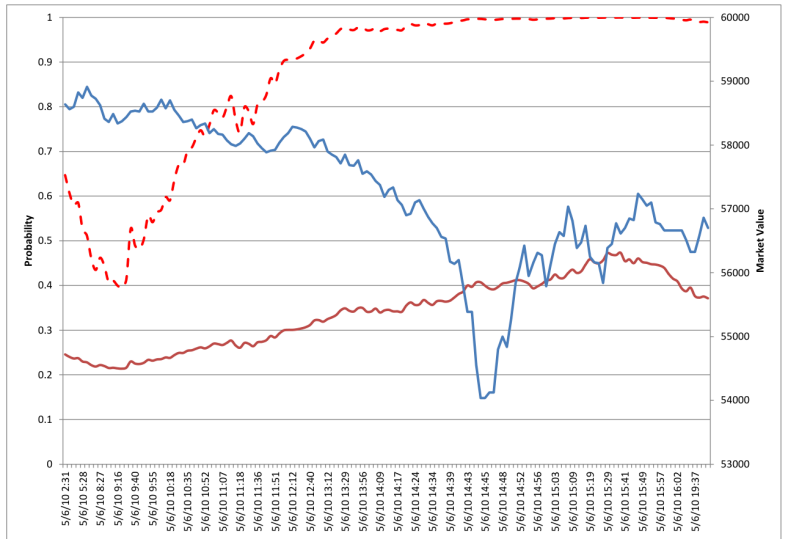
• **VPIN has no training objective.** The paper's only optimisation is the parameter search, and it maximises the statistic it then reports, on the same data. We select on 2009–11 and evaluate on 2011–26.

Data and our replication setup

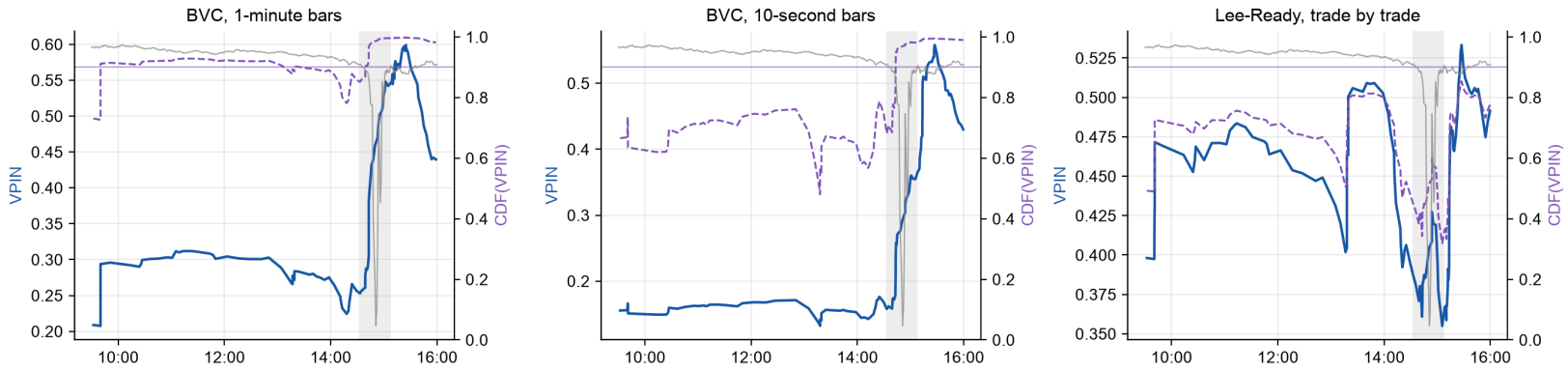
Set-up	Paper	Ours
Market	E-mini S&P 500 and WTI crude futures	IVE (S&P 500 Value ETF) ticks with quotes; BTCUSDT for extensions
Period	Jan 2008 – Aug 2011 (incl. 2008 crisis)	2009-09-28 – 2011-08-15; out of sample to Sep 2026
Trading hours	almost 24 hours	09:30–16:00 New York
Classification	bulk, one-minute bars	same, plus 10-second bars and Lee-Ready
Settings	V = ADV/50, n = 50, whole-sample σ and CDF	identical

- Main deviations: ETF instead of futures; regular hours only; our sample starts in Sep 2009 (no 2008 crisis); 42 IVE trades at prices like $0.08 that the exchanges **cancelled** on 6 May 2010 are removed. 29% of IVE one-minute bars have no price change, and bulk classification splits them 50/50.
- **±** on our numbers: block bootstrap over calendar months (1,000 replications), because VPIN windows overlap and volatility clusters.

Result 1: the flash crash, paper vs ours



Paper, Fig. 4(a): E-mini, one-minute bars



Ours: IVE, 6 May 2010, three classification schemes

- **One-minute bars:** CDF > 0.9 from 09:40, **4.9 h** before the fall, as the paper says.
- **10-second bars:** first above 0.9 at 14:43, during the fall. **Lee-Ready:** never (max CDF 0.85).
- One event: no ± possible. The warning depends on the classification choice.

Result 2: the volume clock and the stability of VPIN

Paper: E-mini, 2008–11

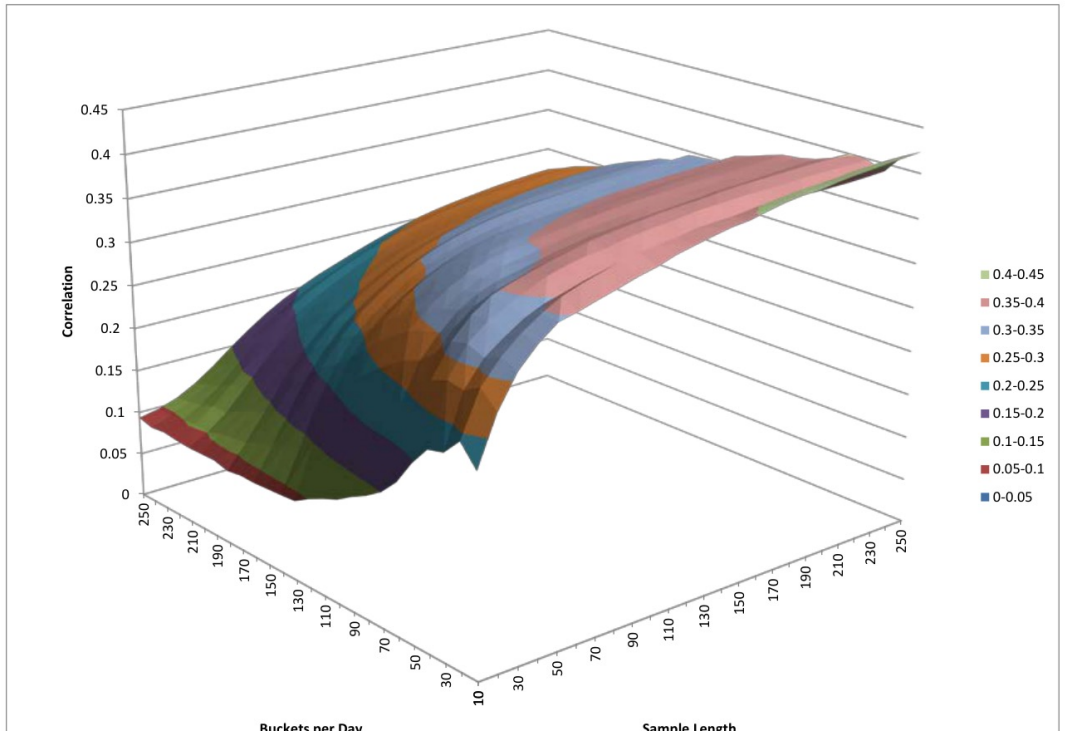
50 draws a day	Clock	Volume
Kurt	34.25	**20.52**
J-B*	40.69	**12.82**
L-B*	40.73	**24.84**
White*	0.10	**0.04**
Max	20.83	**12.26**

Ours: IVE, 2009–11 (± s.e.)

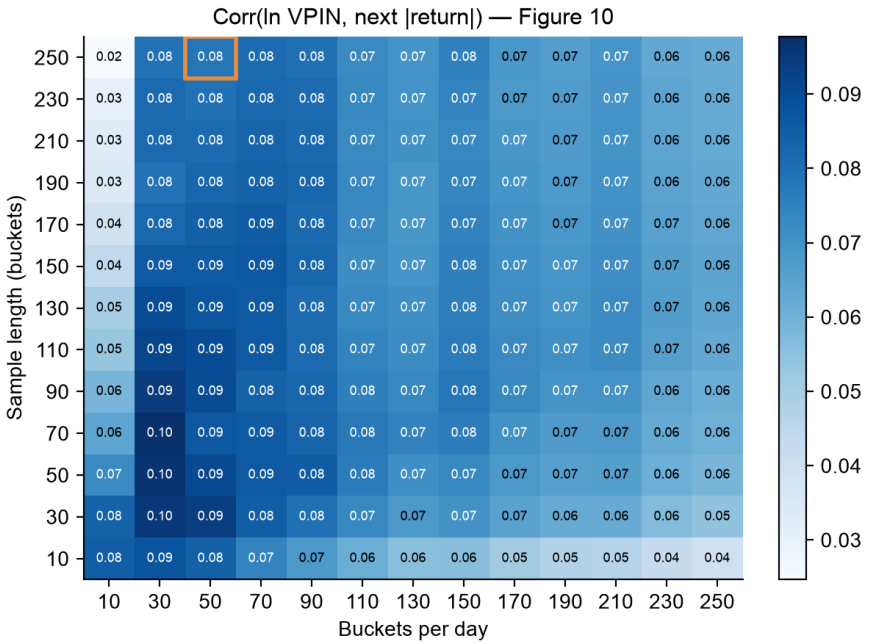
50 draws a day	Clock	Volume	P(volume lower)
Kurt	13.12 ± 3.96	9.05 ± 1.81	0.947
J-B*	4.27 ± 3.11	1.53 ± 0.89	0.947
L-B*	23.13 ± 32.34	43.03 ± 25.23	0.313
White*	0.28 ± 0.10	0.16 ± 0.06	**0.995**
Max	13.08 ± 2.29	9.86 ± 1.51	**0.977**

- Ours: volume time is **significantly** lower on 1 of the 4 test statistics at 50 draws a day and 2 of 4 at 100; tails and White* clearly improve, autocorrelation (L-B*) does not at 50.
- **Stability (Fig. 5):** restarting the buckets 0, 1, …, 999 bars later changes VPIN by 3.7% of its mean on IVE; the paper reports 6.5% for the E-mini.

Result 3: does VPIN predict the next bucket's volatility?



Paper (E-mini): 0.40 at (50, 250), 95% CI [0.392, 0.408] (iid)



Ours (IVE): 0.08 ± 0.05 at (50, 250), 95% CI [0.04, 0.22]

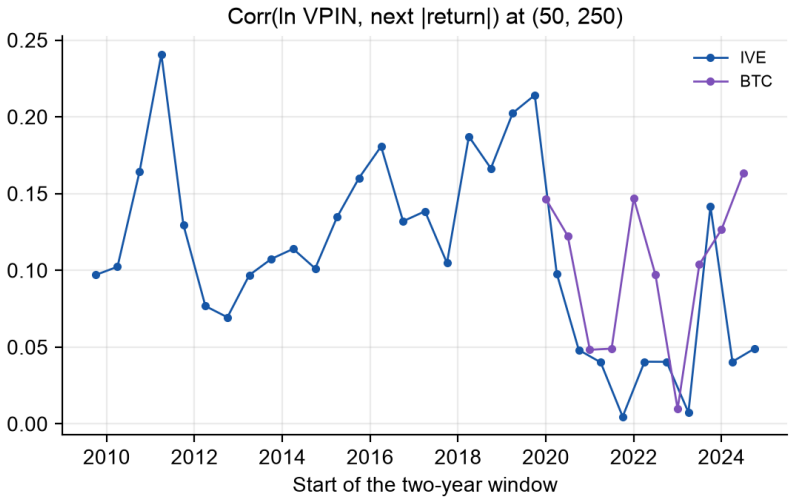
On IVE

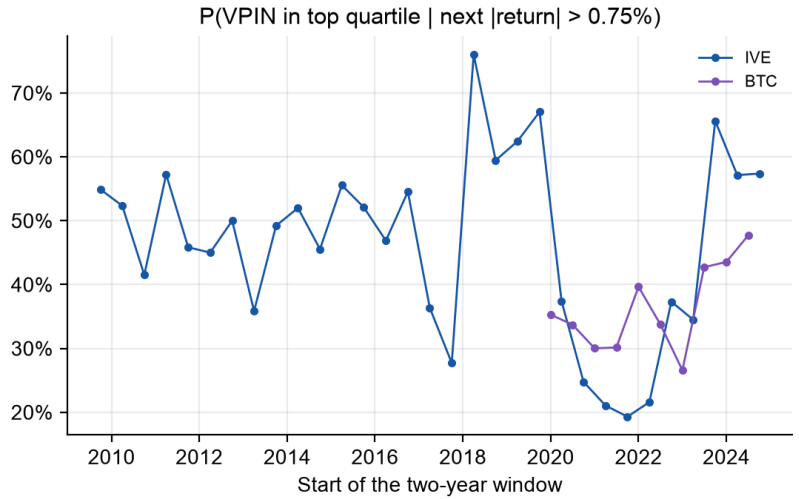
Same sign; 0.40 is outside our interval.

About the ±

An iid interval would give ours as [0.07, 0.09]: overlapping windows make it far too narrow.

Result 4: the paper's predictive claims on every two-year window





- IVE: corr 0.00 to 0.24; P(top quartile | big move) 19% to 76%.
- BTC: corr 0.01 to 0.16; P(top quartile | big move) 27% to 48%. Big moves are routine on BTC, so low VPIN scores as many true positives.

Paper reports (E-mini, 2008–11)

- corr 0.40; P(top quartile | big move) > 84%

Our reading

- No window on either market reaches the paper's numbers.
- Our reproduction could detect any correlation ≥ 0.13 (80% power): the gap is not a matter of power.

Reproduction scorecard

Paper reports: E-mini, 2008–11

#	Claim	Reported
1	VPIN does not depend on where the bucket sequence starts (std / mean across 1,000 starts)	6.5%
2	Volume time is closer to normal than clock time: Kurt, J-B*, L-B*, White* lower	4 of 4 at 50 and at 100 draws a day
3	CDF(VPIN) > 0.9 at least 2 h before the flash crash: BVC, one-minute bars	≥ 2 h
4	... BVC, 10-second bars	≥ 2 h
5	... Lee-Ready, trade by trade	≥ 2 h
6	Low VPIN (bottom quartile) → next |return| < 0.25%	> 90%
7	Corr(ln VPIN, next |return|) at (50, 250)	0.40 [0.392, 0.408]
8	P(VPIN in top quartile | next |return| > 0.75%) at (50, 250)	> 84%
9	P(move > 0.75% while VPIN stays in the top quartile) at (50, 250)	51.8%
10	... at (10, 350)	78.6%

Ours: IVE, 2009–11

#	Ours (± s.e.)	95% interval	Verdict
1	3.7%	–	reproduced
2	1 of 4 / 2 of 4 significant	P ≥ 0.95	partly
3	4.9 h	–	reproduced
4	only during the fall	–	not reproduced
5	never (max CDF 0.85)	–	not reproduced
6	96.1% ± 0.5%	[95.0%, 97.2%]	reproduced
7	0.08 ± 0.05	[0.04, 0.22]	weaker
8	51% ± 13%	[19%, 71%]	weaker
9	34% ± 4%	[26%, 43%]	weaker
10	60% ± 12%	[37%, 85%]	inconclusive

How could these results be overfit?

In the paper

• The CDF of VPIN is built from the **whole sample**, future included.

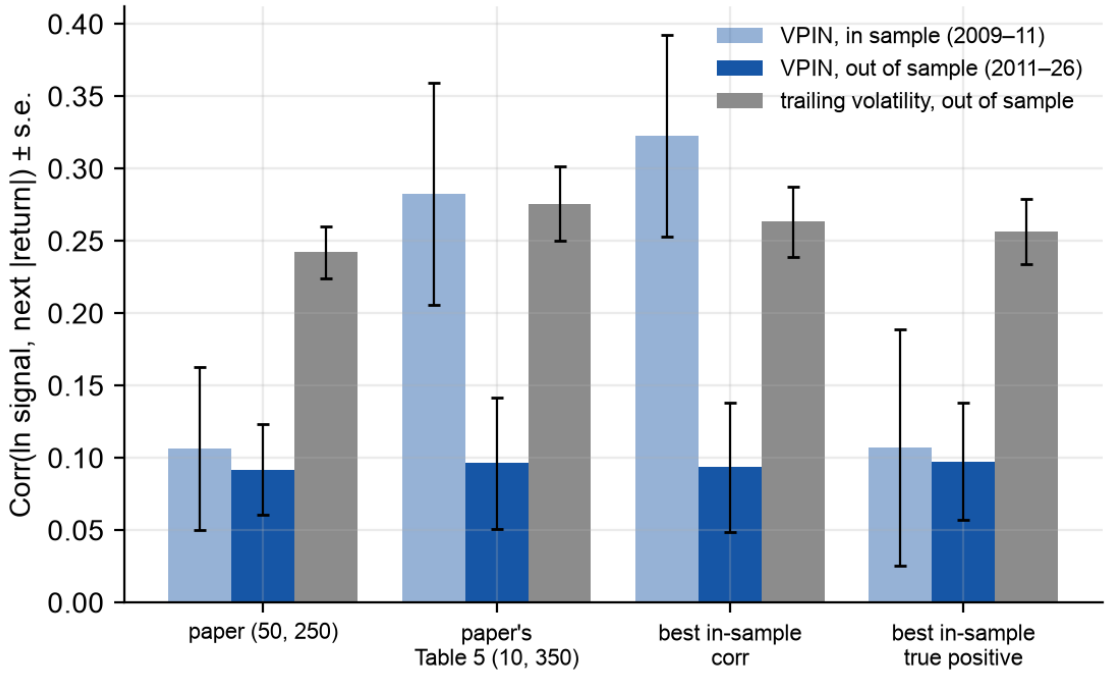
• Bucket size and window are picked by **scanning grids** on the same data (Figs. 10, 12, 14); Table 5 reports the best cell.

• **No out-of-sample** period; its ± assumes independent buckets. One flash crash = one observation.

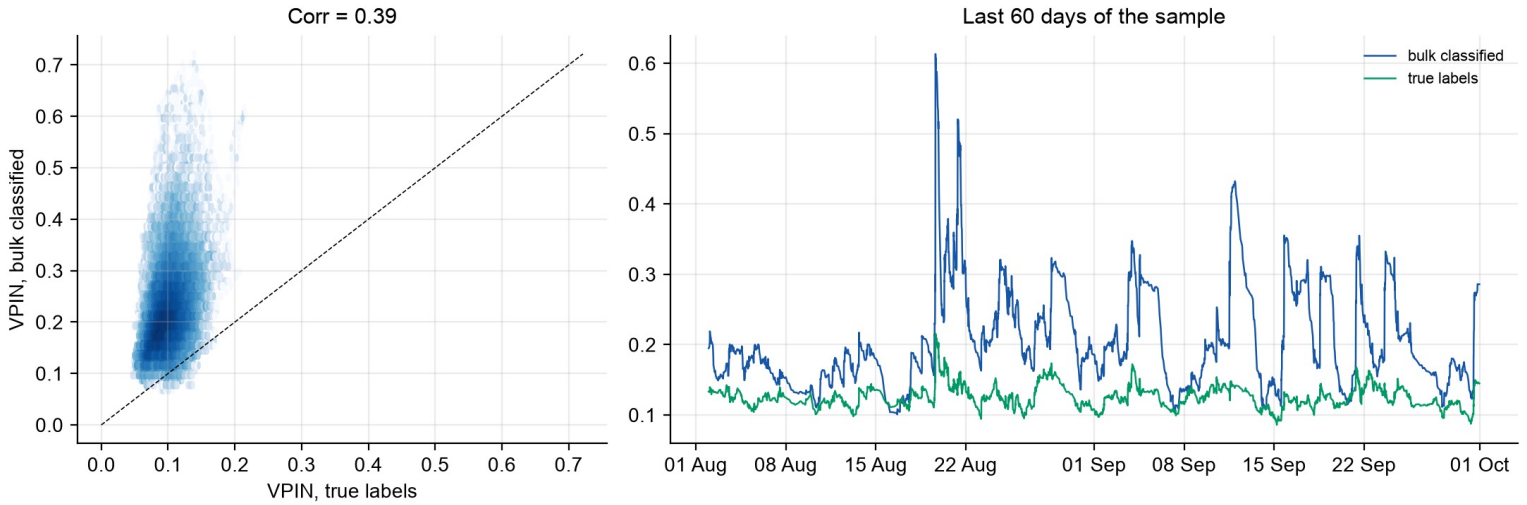
Our check

• Real-time VPIN: σ, V and CDF from **past data only**; parameters frozen in 2009–11, tested on 2011–26.

• Best in-sample true-positive pair (20, 500): **92% ± 5%** → **45% ± 3%** out of sample (bottom quartile: 47% ± 4%).



Extension 1: is bulk classification right? A ground-truth test



- Bars: BVC gets the sign right for **82.1% ± 0.2%** of volume.
- Buckets: BVC inflates |imbalance| **2.2×**; the two VPINs correlate only **0.39 ± 0.04**.

On 5 days with every trade

- Tick rule: **93%–97%** of volume classified right.
- Bucket imbalance vs truth: tick rule **≥ 0.99**, BVC **0.51–0.74** (1-min bars).
- Ranges are across the 5 days. Caveat: Binance aggregates the fills of one order, which helps the tick rule.
- **The paper's claim that bulk beats trade-by-trade classification does not hold here.**

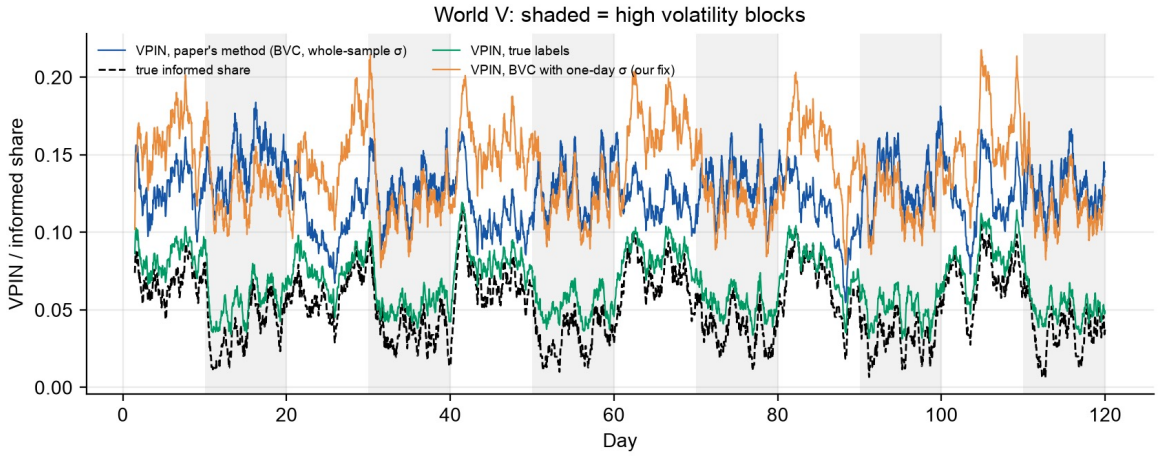
Extension 2: is VPIN more than volatility?

BTCUSDT 2020–26, 50 buckets/day, n = 50	Corr with next |return|	P(top quartile | move > 0.75%)	Time in top quartile	Lift
VPIN (bulk classified)	0.13 ± 0.02	41% ± 3%	27%	1.53 ± 0.08
VPIN (true labels)	-0.12 ± 0.02	17% ± 3%	29%	0.61 ± 0.08
RV (trailing volatility)	0.36 ± 0.03	40% ± 7%	12%	3.31 ± 0.28

- VPIN from **true labels** is negatively related to the next move; bulk-classified VPIN is positive; **trailing volatility wins**.
- BVC imbalance moves with the bucket's own |return| (0.44 ± 0.01 vs 0.29 ± 0.02 for the truth): BVC builds volatility into VPIN.
- Regression with volatility and trade-intensity controls: VPIN adds ΔR^2 = **0.0010 ± 0.0005** on BTC, **0.0025 ± 0.0011** on IVE out of sample.

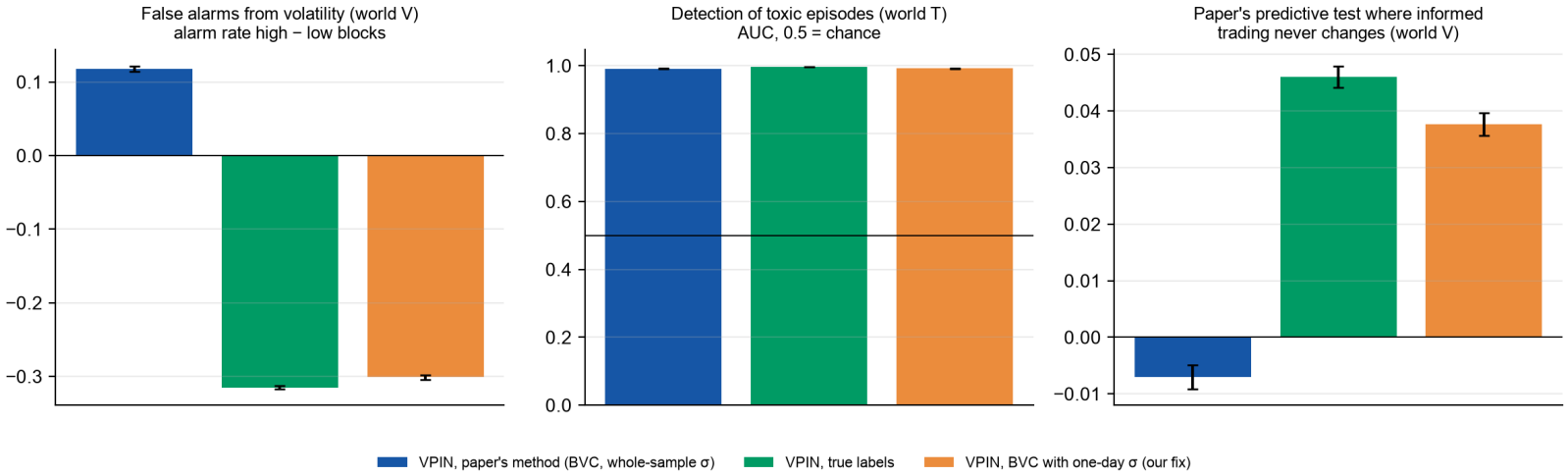
Extension 3: a known-answer test of VPIN — design (pre-registered)

- **Gap in the paper:** its Monte Carlo (Appendix A.2) uses correctly labelled trades and no prices; its VPIN uses bulk classification, which needs prices.
- **Our model:** the paper's PIN model, plus a price: Δp = λ (buys − sells) + public noise. Informed flow moves the price; public noise does not involve any trader.
- **Worlds:** T = more informed trading in some blocks; V = more volatility and uninformed volume, informed trading unchanged; TV = both; 0 = neither.
- **Same pipeline**, truth known, 200 replications per world, rules fixed before running.



World V: the paper's VPIN (blue) stays high in volatile blocks while the true informed share (dashed) falls.

Extension 3: results of the known-answer test



Tracking of the true informed share, world V

- paper's VPIN **0.03**
- true labels **0.96**
- one-day σ (fix) **0.83**

• When volatility changes, the paper's VPIN stops measuring informed trading.

Rule	Pre-registered statement	Result	Verdict
D1	true labels track informed trading	0.98	**holds**
D2	paper's VPIN: false alarms from volatility	+0.118 ± 0.002	**holds**
D3	paper's predictive test fooled by volatility	-0.007 ± 0.001	**does not hold**
D4	one-day σ: fewer false alarms, same detection	-0.301; AUC 0.992	**holds**

Extension 4: the fix on real data

Market (real-time, 50 buckets/day)	Signal	Corr with next |return|	Lift
BTC	VPIN, standard BVC	0.127 ± 0.020	1.53 ± 0.08
BTC	**VPIN, local-sigma BVC (fix)**	0.051 ± 0.017	1.27 ± 0.07
BTC	trailing volatility	0.359 ± 0.032	3.31 ± 0.28
BTC	VPIN, true labels	-0.123 ± 0.024	0.61 ± 0.08
IVE 2011-26	VPIN, standard BVC	0.092 ± 0.031	1.57 ± 0.20
IVE 2011-26	**VPIN, local-sigma BVC (fix)**	0.007 ± 0.023	1.30 ± 0.24
IVE 2011-26	trailing volatility	0.242 ± 0.018	2.76 ± 0.28

Closer to the truth (BTC)

• corr with true-label VPIN: fix **0.47** vs standard 0.39 (+0.08 ± 0.02)

Less volatility in disguise

• its correlation with the next move collapses: the paper's predictive power came from the bias

What remains beyond volatility

• IVE: t = 8.3, $\Delta R^2 = 0.0029$; BTC: negative (t = -4.3)

Inference: placebo, stability, power and every test

Placebo: VPIN with the direction of every imbalance shuffled

Sample	Corr, real VPIN	Corr, direction shuffled	Kept
IVE 2009-11, paper's method (50, 250)	0.078	0.078 ± 0.002	101%
IVE 2011-26 (50, 250)	0.092	0.090 ± 0.001	98%
BTC 2020-26 (50, 50)	0.127	0.128 ± 0.001	101%

- **The direction of trading carries almost none of VPIN's predictive power:** the size of price moves does.
- **Leave one year out:** 0 sign flips across 5 signal–sample pairs.
- **Power:** the reproduction detects any correlation ≥ 0.13; the paper's 0.40 would not be missed.

Ledger of every test

- 26 tests behind our claims
- supported after Benjamini–Hochberg: **17**
- after Bonferroni: **13**
- significant but on the wrong side: **3** (reported as not supported)
- New tests pre-registered before running; every later change logged.

What our work adds to the paper

• **1. Closes the paper's validation gap.** Its Monte Carlo never runs bulk classification. Our known-answer simulation of its own model shows that VPIN, as the paper computes it, stops measuring informed trading when volatility changes.

• **2. Locates where VPIN's predictive power comes from.** With true labels VPIN does not predict volatility (BTC); with shuffled directions it predicts as well as the real VPIN (IVE, BTC). It is the size of price moves, not toxicity.

• **3. A fix that works where it can be checked.** Measuring price changes against the previous day's volatility makes VPIN track true toxicity better on BTC, and removes the volatility artefact.

• **4. An honest evaluation.** Real-time, out of sample 2011–2026, all 41 two-year windows, pre-registration, ± on every metric, and a ledger with multiple-testing control. The paper's predictive numbers do not appear anywhere.

Confirms on new data: Andersen & Bondarenko (2014, 2015); Chakrabarty, Pascual & Shkilko (2015).

Critique and next steps

Critique

• Bulk classification **builds volatility into VPIN**; with true labels the predictive link disappears.

• Whole-sample CDF and grid-searched parameters: **look-ahead and selection** with no out-of-sample test.

• Main evidence is **one event** (6 May 2010); on IVE the warning depends on bar size and classifier.

• No baseline: the paper never compares VPIN with **plain volatility**, which wins in our tests.

• Its **±** treats overlapping buckets as independent: far too narrow.

• The authors hold a **patent** on VPIN and disclose a financial interest.

With three more months

• E-mini futures with **exchange aggressor flags** (e.g. CME MDP 3.0): the paper's market, with ground truth.

• VPIN as a **filter for a market-making strategy**: does it reduce adverse-selection losses after costs?

• Correct the grid search for **multiple testing** (White's reality check, deflated statistics).

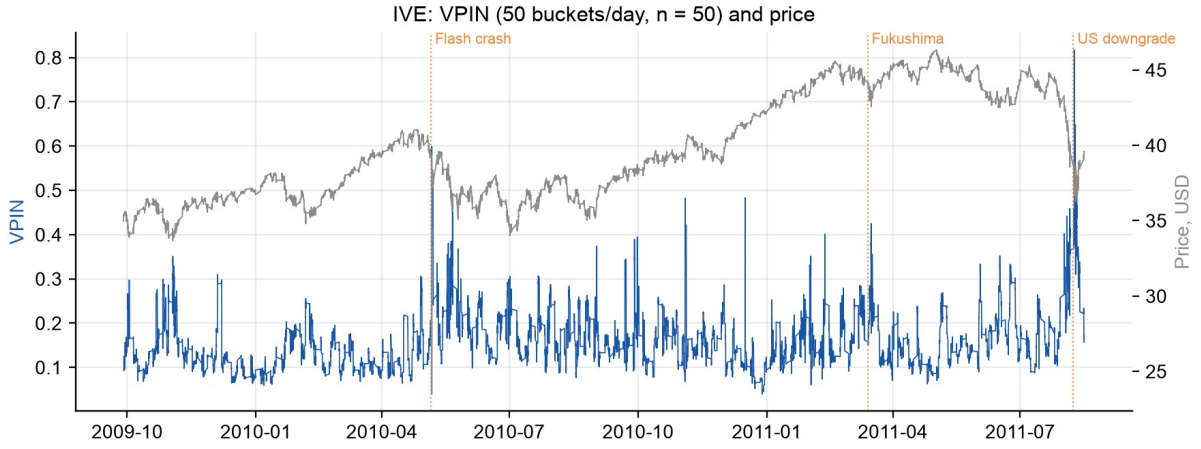
• Combine with **HAR volatility forecasts** and test VPIN's incremental value out of sample.


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Backup: the VPIN metric and how it is read

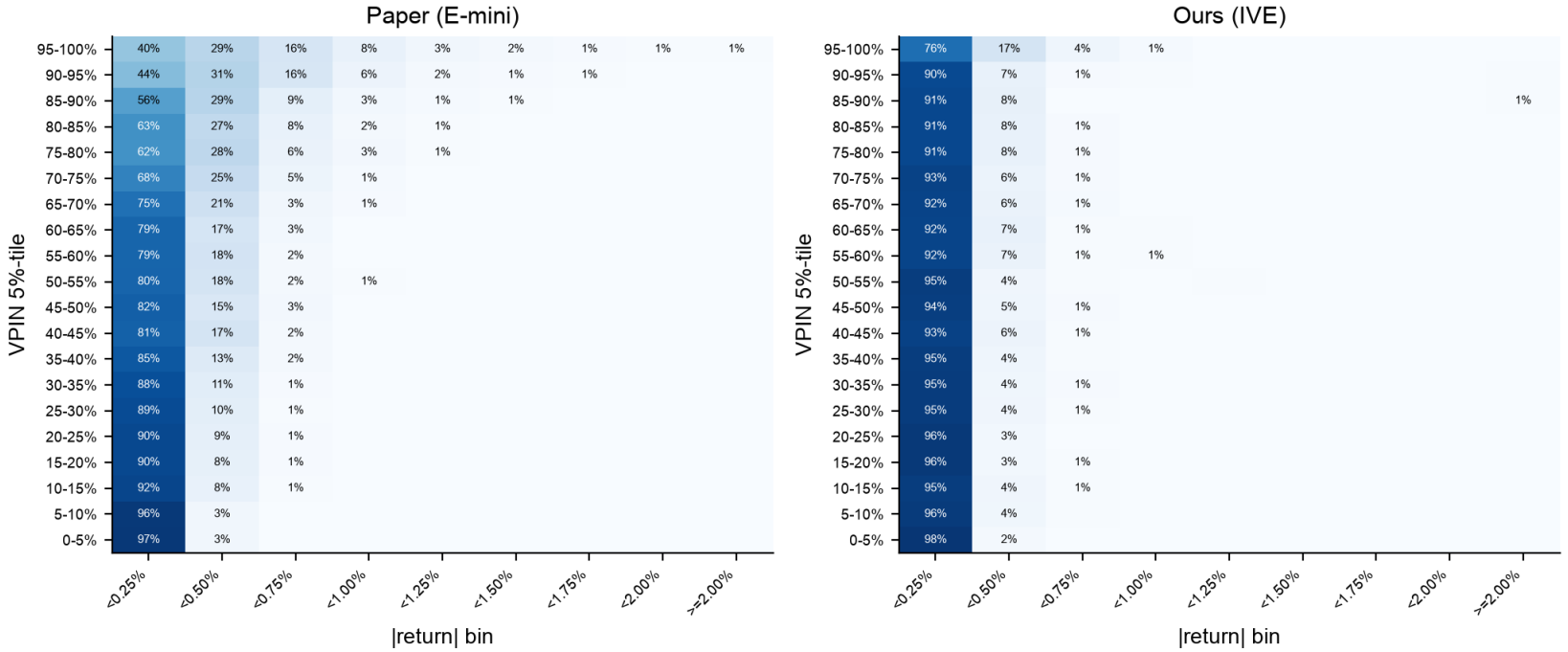
$$\mathrm{VPIN}_\tau = \frac{\sum_{j=\tau-n+1}^{\tau} \left| V_j^S - V_j^B \right|}{nV}$$

- Bucket size **V = average daily volume / 50**; window **n = 50** buckets ≈ one day.
- Updated after **every bucket**: on busy days many times per hour.
- Levels depend on classification and settings, so the paper reads **CDF(VPIN)** over the whole sample. Alarm: CDF > 0.9.
- Claim: high VPIN warns of **toxicity-induced volatility**.



Backup: high VPIN and large moves (Tables 4 and 5)

Table 4a: P(next |return| bin | VPIN 5%-tile), (50, 250)



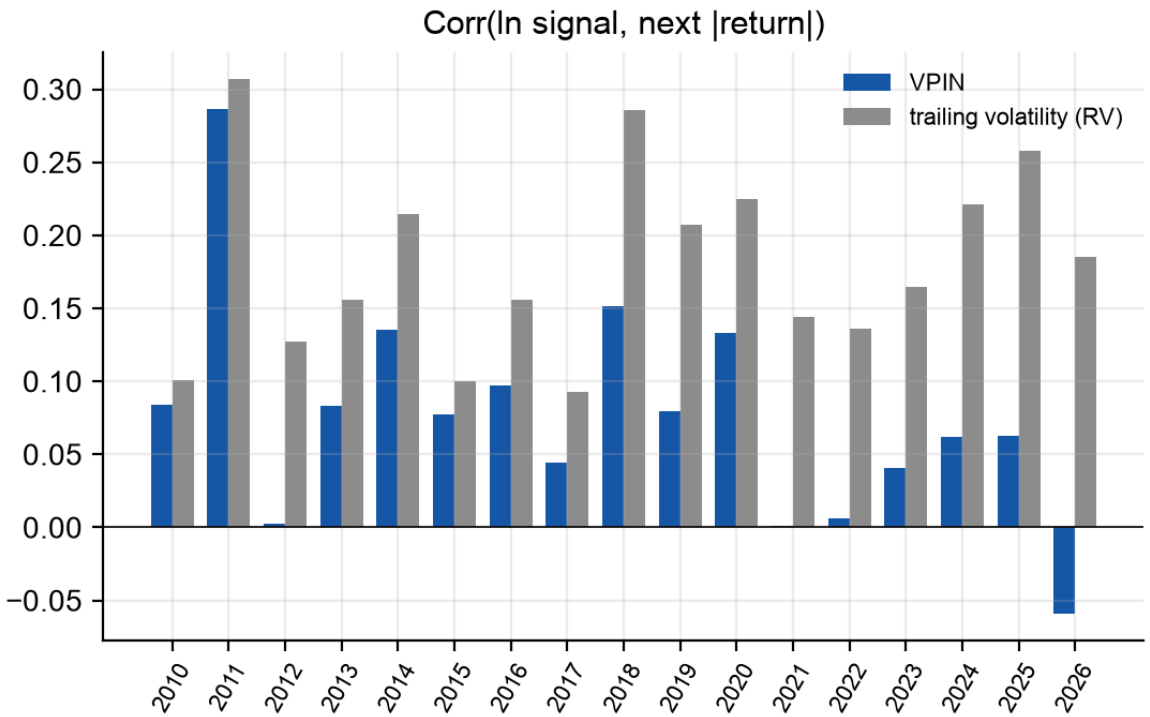
Paper reports (E-mini)

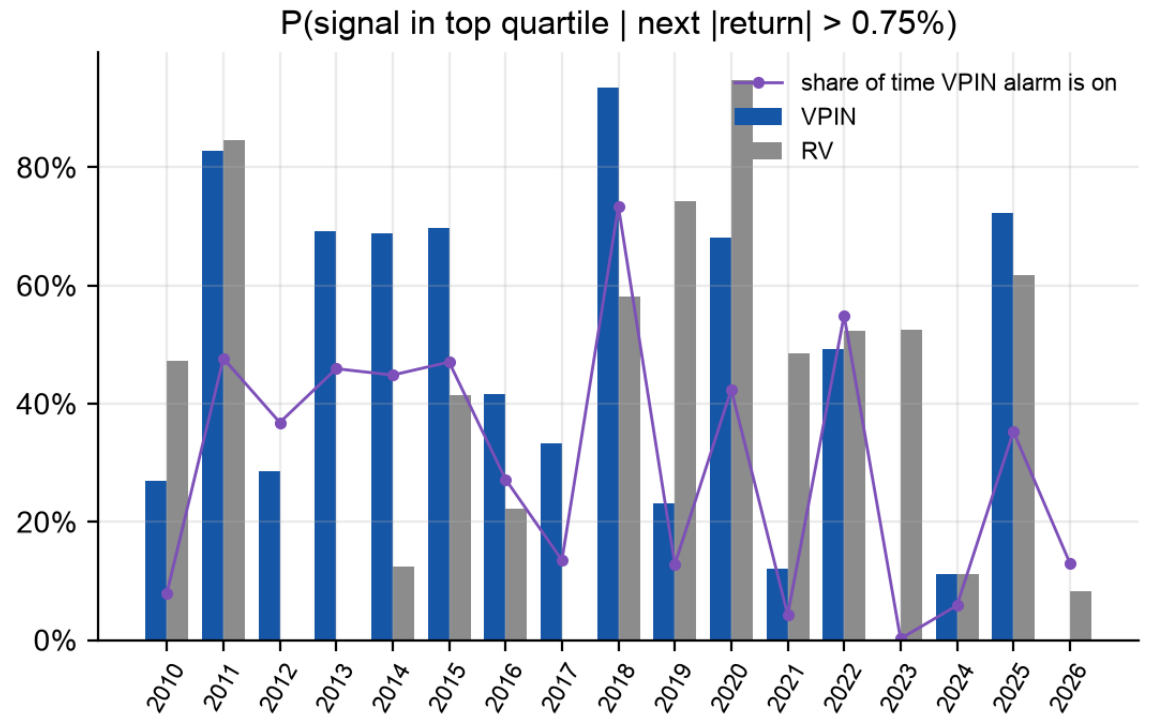
- Bottom-quartile VPIN → next |return| < 0.25%: > 90%
- P(VPIN top quartile | move > 0.75%): > 84%
- Big move while VPIN stays high, (10, 350): 78.6%

On IVE (± s.e.)

- Bottom quartile → < 0.25%: **96.1% ± 0.5%**
- P(top quartile | move > 0.75%): **51% ± 13%**
- Big move while VPIN stays high, (10, 350): **60% ± 12%**, 95% CI [37%, 85%]
- **Low VPIN = calm reproduces; the warning value is weaker or uncertain.**

Backup: real-time VPIN out of sample, year by year





• Paper's (50, 250), 2011–26: corr VPIN **0.09 ± 0.03** vs trailing volatility **0.24 ± 0.02**; lift **1.57 ± 0.20** vs **2.76 ± 0.28**.

Backup: crashes after the paper

Market	Day	Largest 1-hour fall	VPIN alarm	VPIN alarm, true labels	Volatility alarm
BTC	2021-05-19	-22.9%	on	off	off
BTC	2024-08-05	-8.9%	off	off	off
BTC	2025-10-10	-11.6%	off	on	off
IVE	2011-08-08	-3.2%	on	–	on
IVE	2015-08-24	-5.1%	on	–	on
IVE	2018-02-05	-2.7%	on	–	off
IVE	2020-03-16	-4.6%	on	–	on
IVE	2024-08-05	-1.0%	on	–	off

	Alarm	Hits	On overall	p
BTC	VPIN	1 of 3	10%	0.261
BTC	VPIN, true labels	1 of 3	14%	0.355
BTC	Volatility	0 of 3	3%	1.000
IVE	VPIN	5 of 5	15%	< 0.001
IVE	Volatility	3 of 5	9%	0.006

• p: chance of that many hits if the alarm were on at random with its overall rate (falls treated as independent).

• IVE: VPIN on at every fall. BTC: nothing reliable. Few events.

Backup: repository and how to run it

```
src/vpin/
  data.py          IVE ticks, BTC klines and trades; cancelled trades
  bars.py          time bars, within-day price changes
  classify.py      BVC, tick rule, Lee-Ready
  buckets.py       equal-volume buckets
  metric.py        VPIN, full-sample and real-time CDF
  analysis.py      Tables 1–5, Figures 10–14
  realtime.py      no-look-ahead pipeline
  paper_numbers.py  the paper's numbers with pages
scripts/  download, reproduce, extensions, report, slides
tests/    unit tests (incl. Table 1 formula check)
DEVIATIONS.md  every deviation and silent decision
```

Run

pip install -r requirements.txt

pytest -q

python scripts/download_kibot.py

python scripts/download_binance.py

python scripts/reproduce.py

python scripts/extensions.py

python scripts/make_report.py

Free data: 942,321 IVE trades in the paper's period; BTCUSDT 2020–2026.